

Data Cleaning & Preprocessing Report

Mbala Municipal Council Web Dataset
CSC 4792 Data Mining and Warehousing — Mini Project
Project Team 17

1. Introduction

This report documents the cleaning and preprocessing applied to the Mbala Municipal Council page-level dataset prepared for the CSC 4792 Data Mining and Warehousing mini-project. The assignment requires a functional codebase covering data extraction, cleaning and preprocessing, with the dataset creation process rigorously documented in a Jupyter Notebook using Markdown cells. The project also requires CSV files to use the pipe character (|) as the separator and to follow the specified db-unza26-csc4792-[DESCRIPTION].csv naming convention.

2. Dataset Before/After Processing

Property	Final value
Council	Mbala Municipal Council
Project team	17
Page-level records	42
Number of columns	8
File type	CSV
Delimiter	Pipe character ()
Primary source	Supplied scraped extract of the Mbala Municipal Council website
Final page-level filename	db-unza26-csc4792-mbala_municipal_council_cleaned.csv

3. Final Dataset Structure

Field	Description	Processing applied
url	Original Mbala Municipal Council page URL.	Missing values filled with empty strings; whitespace normalised.
page_id	Council page identifier, where available.	Converted to nullable integer, then string; missing values retained as blank.
post_id	Council post identifier, where available.	Converted to nullable integer, then string; missing values retained as blank.
title	Title of the page or post.	Missing values filled with empty strings; whitespace normalised.
date	Publication date, standardised where available	Converted with pandas to a standard YYYY-MM-DD representation where conversion
topics	Topic labels associated with the record.	Missing values filled with empty strings; whitespace normalised. Semicolon-separated
text_content	Extracted textual content from the page or post	Missing values filled with empty strings; repeated whitespace collapsed and surroundi
text_length_chars	Number of characters in cleaned text content.	Recalculated directly from text_content after cleaning.

4. Cleaning and Preprocessing Procedures

4.1 Loading and Structural Inspection

The source was loaded as a pipe-delimited CSV using pandas. The dataset structure was inspected before transformation so that the cleaning operations could be applied to the actual page-level fields. The final cleaned dataset contains 42 records and 8 columns.

4.2 Identifier Standardisation

The `page_id` and `post_id` fields were standardised using nullable integer conversion followed by string conversion. This preserves identifier values without treating them as continuous numerical measurements. Where an identifier was not available in the source, the cleaned representation remains blank rather than inventing an identifier.

4.3 Date Standardisation

The date field was processed with pandas datetime conversion using `errors='coerce'`. Where a valid date could be interpreted, it was represented in YYYY-MM-DD format. Values that could not be converted were not replaced with guessed dates.

4.4 Missing-Value Handling

Textual fields (`url`, `title`, `topics` and `text_content`) were filled with empty strings where missing. Identifier fields were allowed to remain blank where the source did not provide an identifier. No missing project values or financial figures were estimated or fabricated.

4.5 Text Normalisation

Repeated whitespace was collapsed to a single space and leading/trailing whitespace was removed from the main text fields. This produces cleaner text for downstream text mining, information retrieval and natural-language analysis while retaining the substantive source content.

4.6 Text-Length Feature Creation

The `text_length_chars` field was recalculated from the cleaned `text_content` field. This ensures that the recorded character count corresponds to the final stored text rather than to an earlier, uncleaned representation.

4.7 Topic Profiling

The `topics` field contains semicolon-separated topic labels. For descriptive profiling, the labels were split and expanded into individual topic observations. This profiling step was used for analysis and did not alter the master page-level records.

4.8 Duplicate and Consistency Checks

Validation included checking record counts, duplicate URLs, duplicate structured record identifiers, missing values and relevant numeric fields. Source URLs were retained to support provenance and future enrichment.

5. Validation

The cleaned dataset was subjected to final validation before export. The workflow checks that the final structure is available for inspection, that text length is derived from cleaned text, and that the output can be read back using the same pipe delimiter. The notebook also contains validation logic for the final dataset structure. The project specification explicitly assesses the correctness of data cleaning and preprocessing as part of the 40-mark Jupyter Notebook component.

6. Export and File Standardisation

The final dataset is exported as a CSV using the required pipe character (`|`) separator and without an additional dataframe index. The intended project storage location is the `data/processed` directory, separating the cleaned analytical dataset from the original raw source.

Final output path:

```
../data/processed/db-unza26-csc4792-mbala_municipal_council_cleaned.csv
```

7. Relationship to the Structured CDF/LGEF Dataset

In addition to the page-level dataset, the project curates a structured CDF/LGEF dataset from explicit project, asset, empowerment and bursary information appearing in the supplied source records. Its fields include `record_id`, `category`, `project_or_programme`, `ward`, `cdf_year`, `amount_kwacha`, `progress_percent`, `beneficiaries_or_units`, `sector` and `source_url`. Numeric amounts, progress percentages and beneficiary/unit counts are checked where

applicable. Only information explicitly supported by the source is retained.

8. Data Quality and Limitations

The resulting dataset is a cleaned and structured representation of the supplied web extract, not a complete archive of all information published by Mbala Municipal Council. Some source pages identify downloadable documents without providing their full contents in the supplied extract. Consequently, the project does not claim to contain complete council budgets, full Integrated Development Plans, complete Ward Development Committee records, or every CDF beneficiary or project. Missing values were not estimated or fabricated.

9. Reproducibility

The cleaning workflow is implemented in Python using pandas and documented in the Jupyter Notebook. The notebook records the loading, cleaning, profiling, validation and export stages so that another project member or reviewer can reproduce the processing sequence. Source URLs are retained as provenance information.

10. Conclusion

The preprocessing workflow converts the supplied Mbala Municipal Council web extract into a consistent, analysis-ready page-level dataset. The principal operations were identifier standardisation, date conversion, missing-value handling, whitespace normalisation, text-length recalculation, topic profiling and validation. The resulting 42-record, 8-column dataset is stored as a pipe-delimited CSV in accordance with the CSC 4792 project specification.

References

- University of Zambia. (2026). CSC 4792 Data Mining and Warehousing Mini Project Practical Assignment, 4 September 2026.
- Mbala Municipal Council. Official website: <https://www.mbalacouncil.gov.zm/>
- Republic of Zambia. Local Government Act No. 2 of 2019; Local Government (Amendment) Acts 2023 and 2026.
- Phiri, L. (2026). A Multi-Source Dataset for CS1 Failure Prediction [Dataset]. Kaggle.